

The Essence of the Cellular Automaton

A self-directed emergent spiral on a pulsating wavefront

1 State per cell

Each cell of the L^3 lattice ($L = 201$ in the reference build; 321 for the full-size run) stores only:

$$\text{cell} = (r^2, r, \text{active}, \text{spin}, w),$$

where r^2 is the squared Euclidean distance to the center (**INF** = not yet visited), r is the integer radius, two bits mark membership — **active** (on the pulsating shell) and **spin** (on the spiral arm) — and $w \in [0, 9L - 1]$ is a fixed per-cell value, hashed once from the coordinates at startup, that feeds the direction election of Sect. 6.

2 Pulsating wavefront — the “sum of odds”

The distance to the center is never computed by a square root or by multiplication: it propagates through BFS. A unit step along axis i changes r^2 by exactly the corresponding odd number, because

$$(n+1)^2 - n^2 = 2n+1. \quad (1)$$

Hence, for a cell x and its neighbor $x + e_i$, where c_i denotes the coordinate centered on axis i :

$$r^2(x + e_i) \leftarrow \min(r^2(x + e_i), r^2(x) + 2|c_i| + 1). \quad (2)$$

On top of this field acts the pulse: a *triangular* wave in r^2 space,

$$p(t) \in [0, 0.92 R_{\max}^2], \quad R_{\max} = \frac{L}{2}, \quad (3)$$

which expands and contracts over time. A cell belongs to the living shell exactly when

$$\text{active}(x, t) \iff |r^2(x) - p(t)| \leq \tau, \quad (4)$$

with a small tolerance $\tau = 1$. The 0.92 factor keeps the whole dynamics strictly inside the lattice: no active cell ever exceeds $r = \lfloor \sqrt{0.92} R_{\max} \rfloor$.

3 The helical walker (3D Bresenham around an arbitrary axis)

3.1 Target geometry

A cylinder of radius $R_{\text{cyl}} \approx 0.182 R_{\max}$ whose axis is the line $\{P + sA\}$, with

$$|A| = \frac{L}{2}, \quad |P| = R_{\text{cyl}}, \quad P \perp A. \quad (5)$$

Because the center lies on that cylinder surface, the helix starts at $r = 0$ and ends at $r = L/2$ (polar angle $\theta : 0 \rightarrow \pi$): one full half-turn while climbing from the core to the outer shell.

3.2 Radius to the axis without multiplication

For $u = v - P$ (offset from the cylinder axis line), the squared distance to the axis is

$$d^2 = \frac{|u \times A|^2}{|A|^2}, \quad (6)$$

so the walker compares $|u \times A|^2$ against the fixed target $\text{TGT} = R_{\text{cyl}}^2 |A|^2$. Both the cross product $c = u \times A$ and its squared norm $E = |c|^2$ are maintained *incrementally*: a lattice move e_m changes c by the constant vector

$$D_m = e_m \times A, \quad (7)$$

and changes E by

$$G[m] = 2(c \cdot D_m) + |D_m|^2, \quad (8)$$

whose own update after move k uses the precomputed table

$$K[k][m] = 2 D_k \cdot D_m. \quad (9)$$

This is the exact generalization of the classic “ $2dx + 1$ ” increment from scalar coordinates to cross-product space.

3.3 Orbital step

Among the six lattice moves whose tangential projection is positive ($t = -c$ enforces a single sense of rotation), pick the one minimizing the radial error

$$|E + G[m] - \text{TGT}|, \quad (10)$$

restricted to the tolerance band $|\cdot| \leq \text{TOL}$ (about one cell); ties favor the largest angular progress. If no move stays inside the band, fall back to minimum radial error.

3.4 Climb step

A multi-dimensional DDA along the axis: each accumulator obeys $dda_i += |a_i|$; the largest one fires the climb move and then subtracts $|a_x| + |a_y| + |a_z|$. Lattice moves are therefore distributed exactly in proportion to the axis components.

3.5 Interleaving

A Bresenham accumulator interleaves orbital and climb steps with period

$$\text{orbit_total} + \text{climb_total}, \quad (11)$$

tracing the helix one cell per tick.

4 The constraint that defines the automaton’s character

At runtime only additions, subtractions, shifts, and comparisons exist: *no multiplication, no division, no floating point, no trigonometry*. Everything that depends on the chosen axis — D_m , $K[k][m]$, P , TGT , TOL , orbit_total — is derived once at initialization time, where multiplication is allowed. All geometric intelligence lives in precomputed constants plus local comparison rules.

5 Emergence

spin draws the arm; **active** paints the living shell. The spiral itself is never programmed: it emerges from the local rule “minimize radial error while keeping the rotation sense,” executed on top of the pulsating bubble.

6 Electing the direction: the ant-view tournament on w

The fused automaton is not told where to spiral. At every expansion limit of the pulse — the exact tick where $p(t)$ turns from ascending to descending — the automaton harvests its own next rotation axis out of the w field, through a tournament that no cell referees.

6.1 Sowing

Every **active** shell cell is seeded with the payload

$$\text{payload}(x) = (\text{score}(x) \ll 24) \mid \text{code}(x), \quad \text{score} = H((w(x) + s) \bmod 9L, \text{code}(x)), \quad (12)$$

where $\text{code}(x)$ is the cell's linear index (low bits) and s is the global seed dial. Because the code occupies the low bits, the payload order is a *strict total order*: ties are impossible, so the winner is unique and independent of any scan order.

6.2 Diffusion — and the flood trap

Each pass lets every cell adopt the greatest payload among its six face neighbors,

$$\text{payload}(x) \leftarrow \max(\text{payload}(x), \text{payload}(x \pm e_i)), \quad (13)$$

alternating sweep directions; $2R + 8$ passes guarantee full coverage of the torus. The subtlety: after enough passes the winning payload has been *copied* across a large region of the grid, down to the lowest linear indices. An index scan for the maximum would therefore decode a *copy* — possibly the corner cell $(0, 0, 0)$, yielding $|m| = \sqrt{3}L/2$ and a nonsense axis. The implementation avoids the trap structurally: the true winner — the *seeded* cell holding the greatest payload — is identified once, at sowing time, when payloads are still unique. The diffusion then remains what it should be: pure transport of the news, never a redefinition of the winner.

6.3 From winner to axis

The winner becomes the momentum vector: the raw displacement from the lattice center to the winning cell,

$$\mathbf{m} = x_{\text{win}} - c. \quad (14)$$

Only its *direction* carries meaning; the magnitude is discarded by rescaling,

$$A = \frac{L/2}{|\mathbf{m}|} \mathbf{m}, \quad (15)$$

so the helix always walks on an axis of length exactly $L/2$. The axis is frozen while a walk is alive; the next election (next expansion limit) prepares a fresh direction for the next run. Nothing is prescribed: the direction is harvested from the CA's own state at the exact instant the wave reaches its limit.

7 This edition: the ant-view broadcast

An additional layer follows the same philosophy. Each cell holds one monotonically growing value ($0 = \text{never reached}$). When the walker computes a new spiral point, that cell receives the current tick as its value. Then, every other tick (sweep direction alternating), every cell inspects *only* its six face neighbors and copies the greatest neighbor value if it exceeds its own:

$$v(x) \leftarrow \max(v(x), v(x \pm e_i)). \quad (16)$$

No cell ever senses beyond its neighborhood, yet the values diffuse as waves that in finite time reach every cell of the L^3 grid — global coverage achieved purely by local moves, in lockstep with the CA tick.

Rendered as faint dust over a sparse sample, the *hue* encodes *which expansion wave* delivered the value — one distinct, coherent colour per breathing cycle,

$$\text{hue} = \left\lfloor \frac{v}{T_{\text{half}}} \right\rfloor \cdot \varphi \bmod 1, \quad \varphi = \frac{1+\sqrt{5}}{2}, \quad (17)$$

where T_{half} is the tick length of one expansion phase of the pulse — while the *brightness* encodes how recently the value arrived, glowing for a bounded youth window and then settling to a dim floor.